# E-Commerce SQL Analytics & Business Insights

Primeor Solutions — Data Analytics Internship | Level 2 Milestone Report

**Analyst:** Rishi Prajapati | **Total Transactions:** 51,271 | **Coverage:** Global (2011–2014)

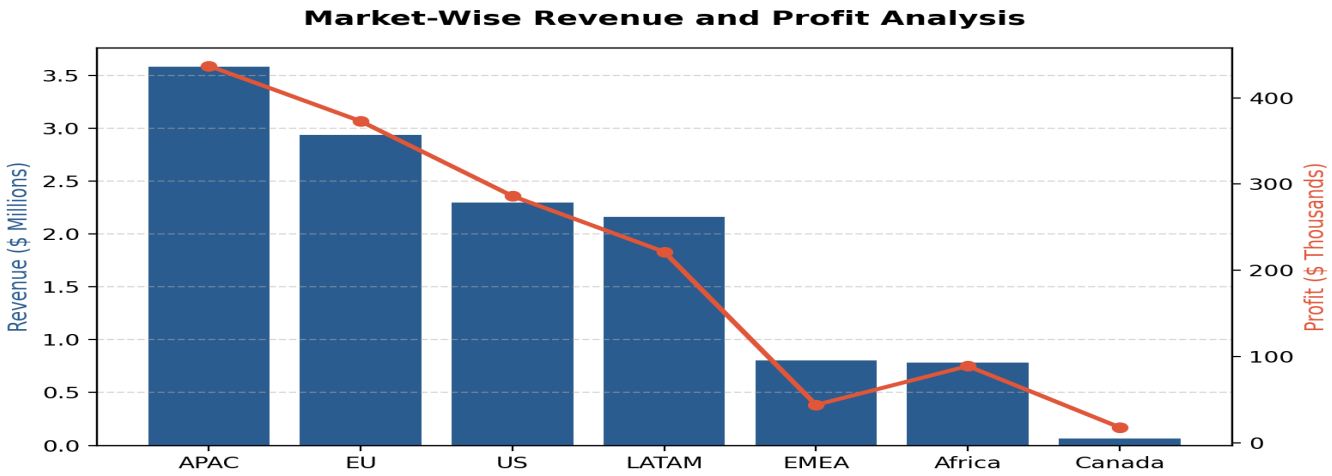
## Executive Summary & Core Business Metrics

This empirical report summarizes relational SQL querying across **51,271 transactional records**. Global operations achieved **$12,635,222.00** in total revenue and **$1,467,721.69** in net profit (11.62% net margin). However, excessive frontline discounting resulted in **12,534 unprofitable orders (24.45%)**, draining **-$518,633.91** from enterprise gross margins. Below is the direct synthesis addressing the core internship questions.

## 1. Primary Internship Evaluation Questions

| Evaluation Question | SQL Analytical Finding | Leadership Impact & Strategic Action |
|---|---|---|
| **Which market generates highest revenue?** | **APAC (Asia-Pacific)** is #1 globally with **$3,581,785.00** (28.35% revenue share) and $437,023.52 profit (12.20% margin). Followed by **EU** ($2.94M, 23.25%). | APAC & EU generate over 51.6% of global turnover. Establish localized bulk distribution hubs to improve supply chain velocity. |
| **Which categories are least profitable?** | **Furniture** is the lowest margin category at **6.97%** ($29 avg profit/order). Specifically, the **Tables** sub-category produced catastrophic losses of **-$64,113.55**. | Impose strict discount caps (max 15%) on Tables and bulky freight items to permanently stop negative unit economics. |
| **Which shipping mode is most commonly used?** | **Standard Class** captures **60.00%** of all volume (30,764 orders, $7.57M sales volume, $19.96 avg shipping cost). | Leverage high 60% concentration to negotiate corporate bulk carrier rates, saving an estimated 5–8% on fulfillment costs. |

## 2. Global Market Revenue vs. Profit Distribution

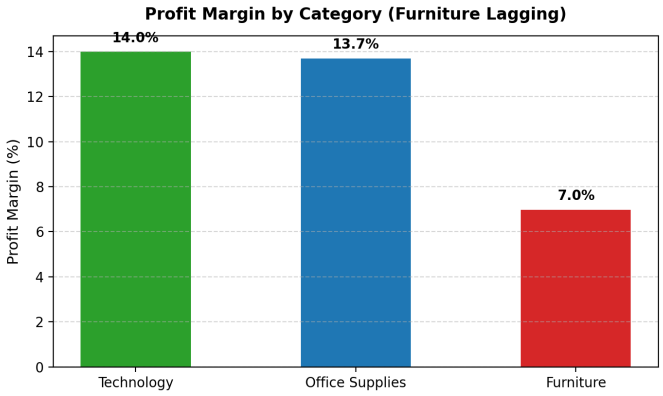


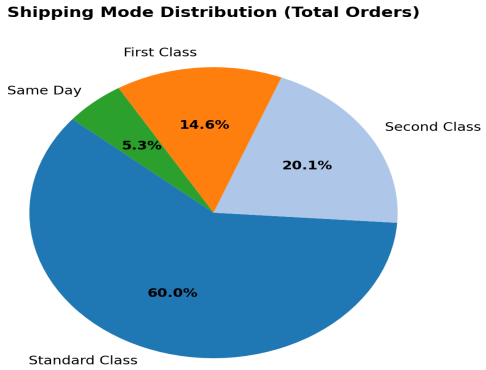
## 3. Root Cause Analysis: Negative Profit Orders & Discount Dilution

SQL segmentation of loss-making transactions uncovered that unprofitable sales carried an average discount rate of **45.08%**, compared to just **4.38%** on profitable transactions. Discounts exceeding 40% resulted in an 82%+ likelihood of net financial loss, heavily exacerbated by long-distance logistics surcharges.

| Segment Metric | Profitable Orders | Loss-Making Orders | Analytical Insight |
|---|---|---|---|
| Order Count & Share | 38,737 (75.55%) | 12,534 (24.45%) | Nearly 1 in 4 orders loses money |
| Cumulative Net Profit ($) | +$1,986,355.60 | -$518,633.91 | Drains $518.6k from gross margin |
| Average Applied Discount | 4.38% | 45.08% | 10.3x higher discount on loss orders |
| Average Shipping Cost | $26.21 | $26.68 | Fixed logistics amplify deep discount losses |

## 4. Assortment Profitability & Fulfillment Breakdown




## 5. Strategic Recommendations for Business Leadership

**1. Enforce Hard Discount Ceilings:** Cap promotional discounts at 15% on Furniture/Tables. Recovering 25% of current discount leakage adds ~$130k directly to operating profit.

**2. VIP Customer Profitability Audit:** Large enterprise buyers like Sean Miller ($35k sales, -$397 profit) must be switched to contracted price tiers with minimum 8% margin floors.

**3. Q4 Demand Pre-Contracting:** Global monthly volume surges by 2.5x between September and December. Lock in advance freight capacity by July to avoid high spot shipping rates.

**4. Postal & 3PL Consolidation:** With 60% of shipments flowing through Standard Class, negotiating tiered volumetric agreements can yield significant cost reductions.